

Generation Interconnection System Impact Study Report

For Queue Project Q12-0159

Blackwater Tie 138 kV

March 27, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0159. The Interconnection Customer proposes a 53.9 MW battery energy storage facility interconnecting at the Blackwater Tie 138 kV substation in the Pennsylvania area. The request was received on September 26, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 113 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,771,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
JACK CREEK to TALENENERGY MARTINS CRK LLC W SHORE 138.0 kV L	LITTLE CHUTE to TALENENERGY MARTINS CRK LLC HARRISBURG 345.0 kV	156.5	238	5.3
SAN LORENZO to AJ SWOPE 138.0 kV	TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Autotransformer	129.4	192	7.2
SAN LORENZO to GRASS CRK 138.0 kV	TALENENERGY MARTINS CREEK LLC HARRISBURG 500-345 kV Transformer	129.1	183	7.2
TALENENERGY MARTINS CREEK LLC W SHORE to AJ SWOPE 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Transformer Bank	122.0	218	7.2
BARNES LAKE to GRASS CRK 138.0 kV	TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Autotransformer	112.1	203	7.2
CURECANTI to MOSER GENERATING STATION 345.0 kV	TOMBSTONE to BIXBY HILL 500.0 kV L	110.3	1,413	5.4
MOUNTAIN 500/345kV TX	LITTLE CHUTE SUBSTATION to FT HUMBUG 345.0 kV	108.4	876	6.8
WESTWOOD GENERATION LLC 345/1kV Auto	FOSTER WHEELER MT CARMEL COGEN 500/345kV Autotransformer	107.3	843	6.4
VICTORIA TC 7001-7007-7008 to BARNES LAKE 138.0 kV	TALENENERGY MARTINS CRK LLC HARRISBURG 500 kV/345 kV Autotransformer	103.5	212	7.2
HILLTOP 500 kV/345 kV Autotransformer	HILLTOP to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500.0 kV L	102.4	1,071	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CARBON CTR 500 kV/345 kV Auto	SHAWVILLE STATION to BLACKROCK 345.0 kV	100.7	660	5.6
LITTLE CHUTE to FOR HUMBUG 345.0 kV	MOUNTAIN to FORT HUMBUG 500.0 kV L	100.6	1,196	8.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JACK CREEK to TALENENERGY MARTINS CRK LLC W SHORE 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,263,000	9% (\$211,000)
SAN LORENZO to AJ SWOPE SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$4,802,000	27% (\$1,300,000)
SAN LORENZO to GRASS CRK 138.0 kV	Rebuild line with higher-capacity conductor	\$3,628,000	27% (\$982,000)
TALENENERGY MARTINS CREEK LLC W SHORE to AJ SWOPE 138.0 kV	Rebuild line with higher-capacity conductor	\$5,894,000	27% (\$1,596,000)
BARNES LAKE to GRASS CREEK SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,576,000	27% (\$427,000)
CURECANTI to MOSER GENERATING STATION 345.0 kV	Rebuild line with higher-capacity conductor	\$28,627,000	1% (\$274,000)
MOUNTAIN 500/345kV TX	Replace transformer with higher-rated unit	\$33,345,000	3% (\$915,000)
WESTWOOD GENERATION LLC 345/1kV Auto	Replace transformer with higher-rated unit	\$10,302,000	12% (\$1,246,000)
VICTORIA TC 7001-7007-7008 to BARNES LAKE 138.0 kV	Reconductor line	\$1,568,000	27% (\$425,000)
HILLTOP 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$18,216,000	1% (\$260,000)
CARBON CTR 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$40,201,000	1% (\$358,000)
LITTLE CHUTE to FOR HUMBUG 345.0 kV	Reconductor line	\$54,105,000	1% (\$777,000)
	Total allocated to project		\$8,771,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.